

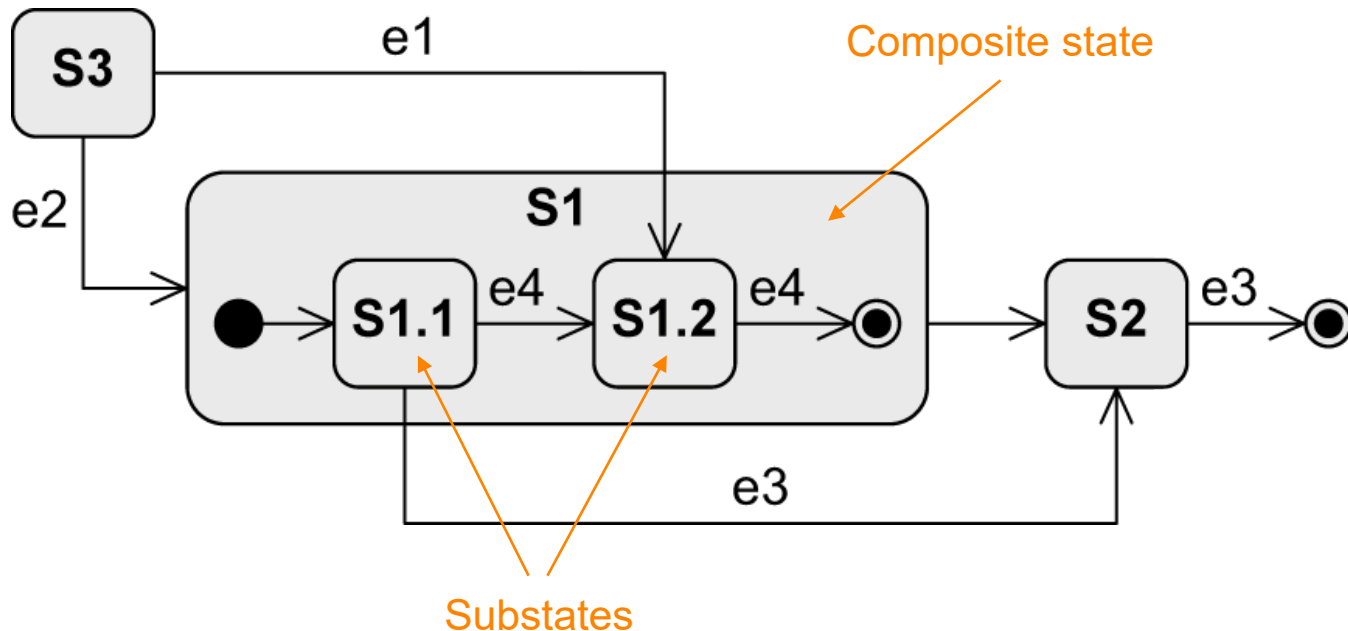
UML State Machines (Advanced)

Software Design (40007) – 2023/2024

Justus Bogner
Ivano Malavolta

Composite state

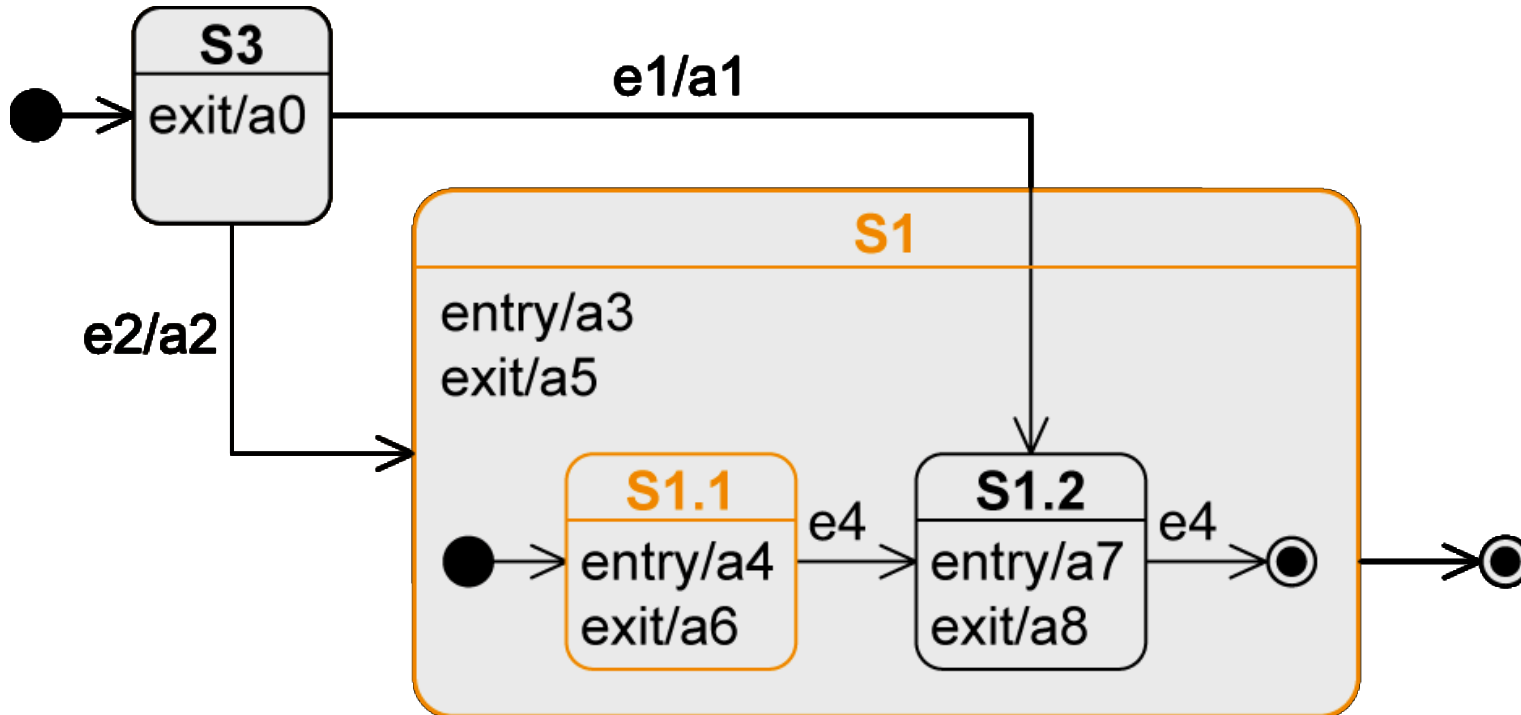
- Contains other states called “substates”
 - Only one of its substates is active at any point in time
- Arbitrary nesting depth of substates allowed (but be careful!)



Entering a composite state (1/2)

- Transition to the boundary
 - Initial node of composite state is activated

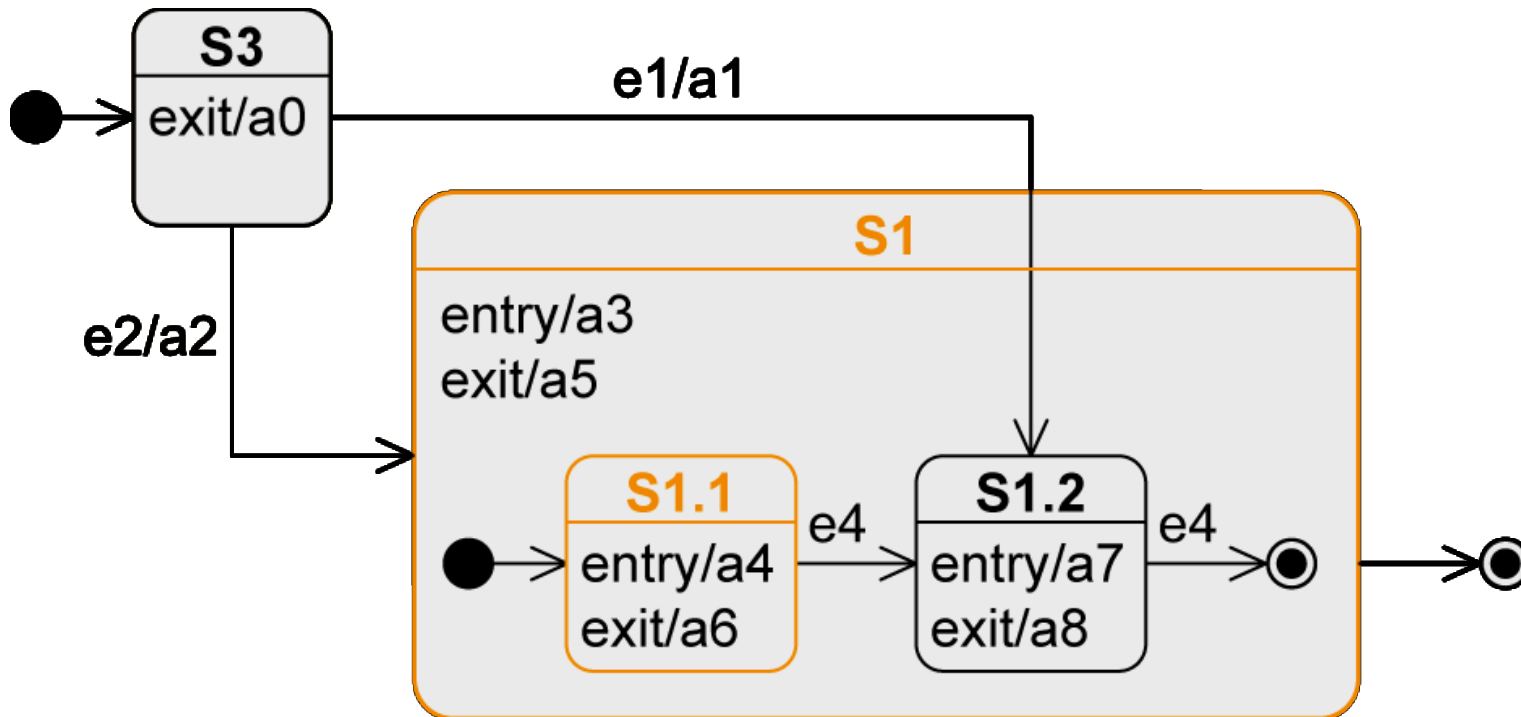
Event	State	Executed Activities
„Beginning“	S3	
e2	S1/S1.1	



Entering a composite state (1/2)

- Transition to the boundary
 - Initial node of composite state is activated

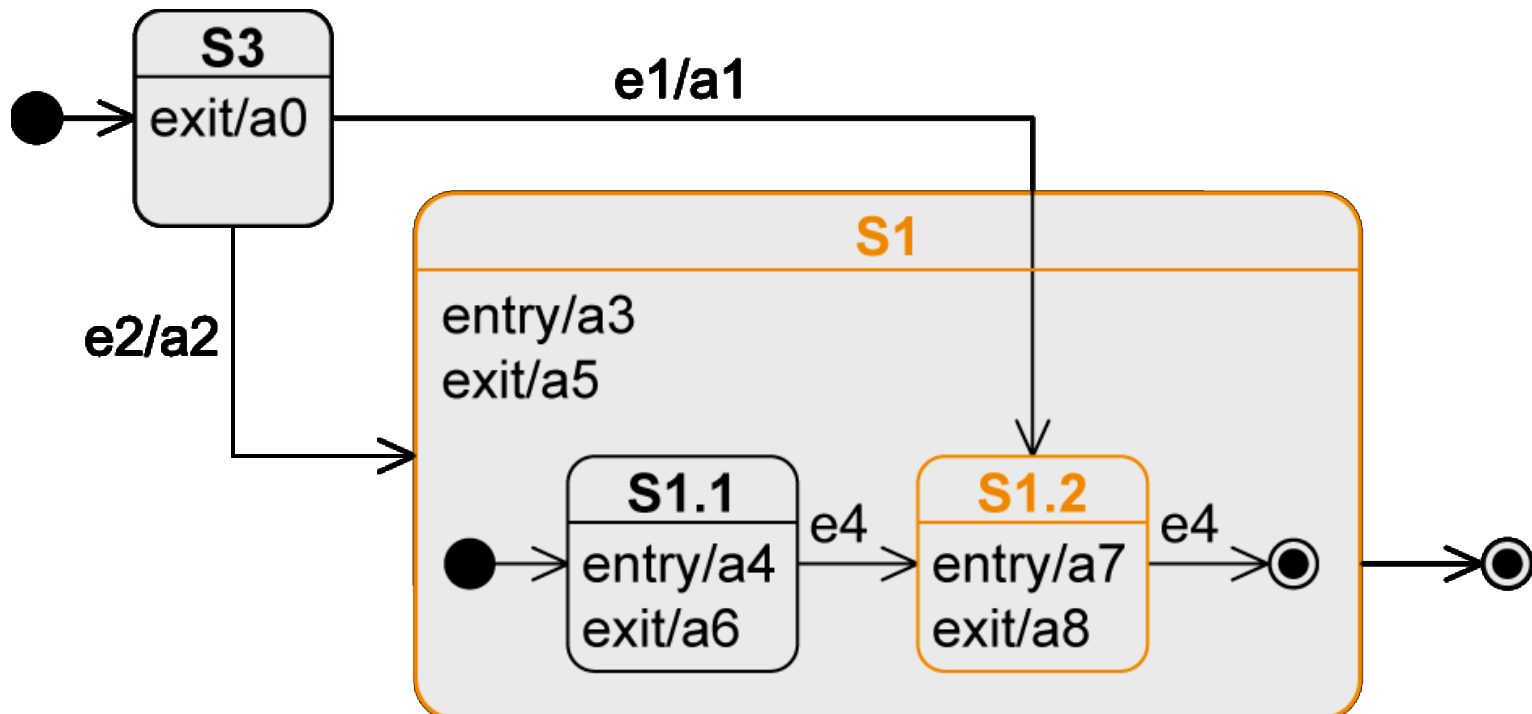
Event	State	Executed Activities
„Beginning“	S3	
e2	S1/S1.1	a0-a2-a3-a4



Entering a composite state (2/2)

- Transition to a substate
 - Substate is activated

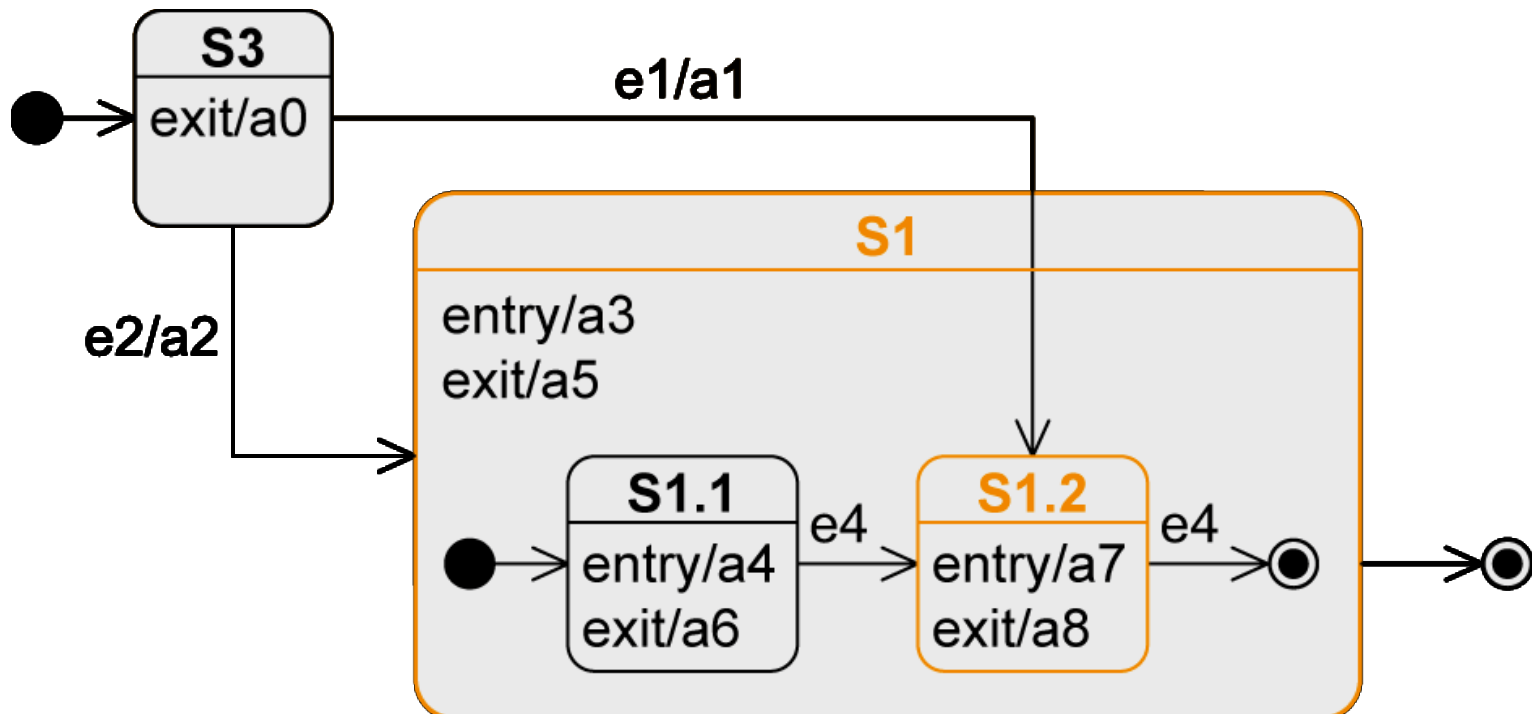
Event	State	Executed Activities
„Beginning“	S3	
e1	S1/S1.2	



Entering a composite state (2/2)

- Transition to a substate
 - Substate is activated

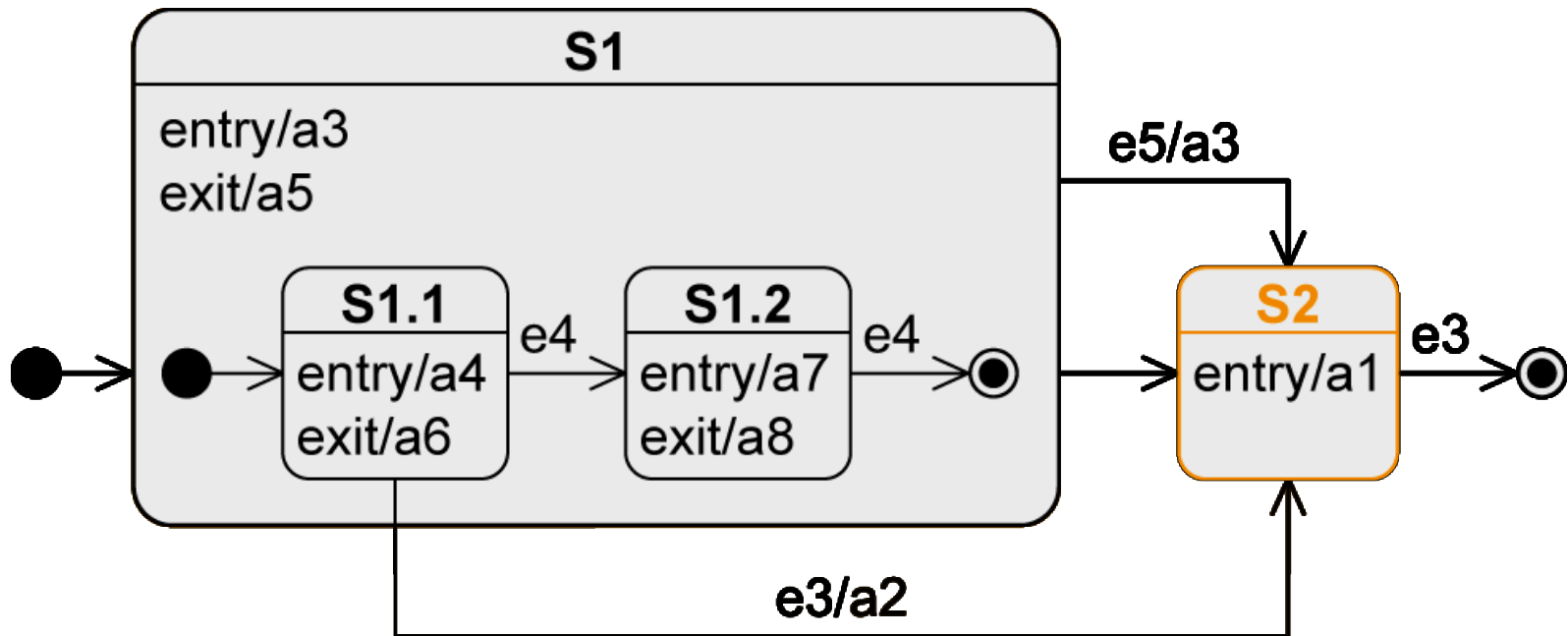
Event	State	Executed Activities
„Beginning“	S3	
e1	S1/S1.2	a0-a1-a3-a7



Exiting from a composite state (1/3)

- Transition from a substate

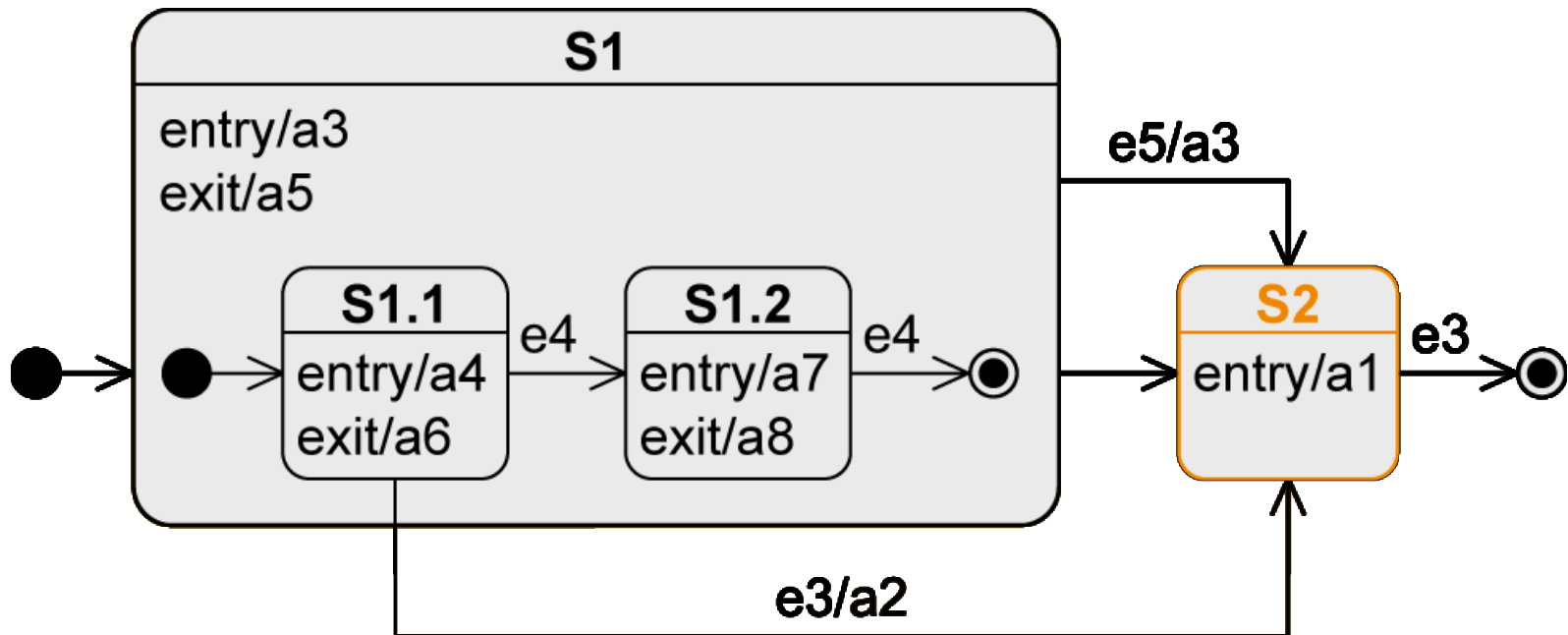
Event	State	Executed Activities
„Beginning“	...	...
e3	...	...



Exiting from a composite state (1/3)

- Transition from a substate

Event	State	Executed Activities
„Beginning“	S1/S1.1	a3-a4
e3	S2	a6-a5-a2-a1

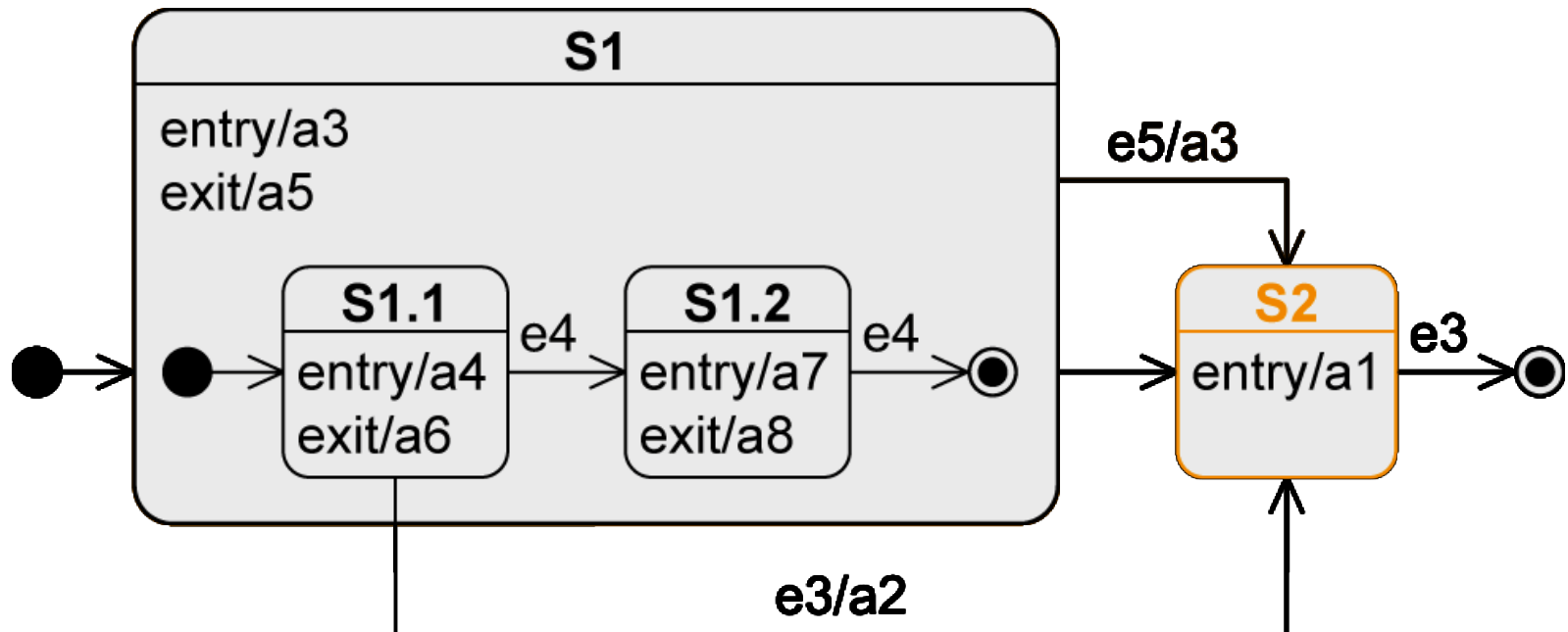


Exiting from a composite state (2/3)

- Transition from the composite state

No matter which substate of S1 is active, as soon as e5 occurs, the system changes to S2

Event	State	Executed Activities
„Beginning“	...	...
e5	...	...

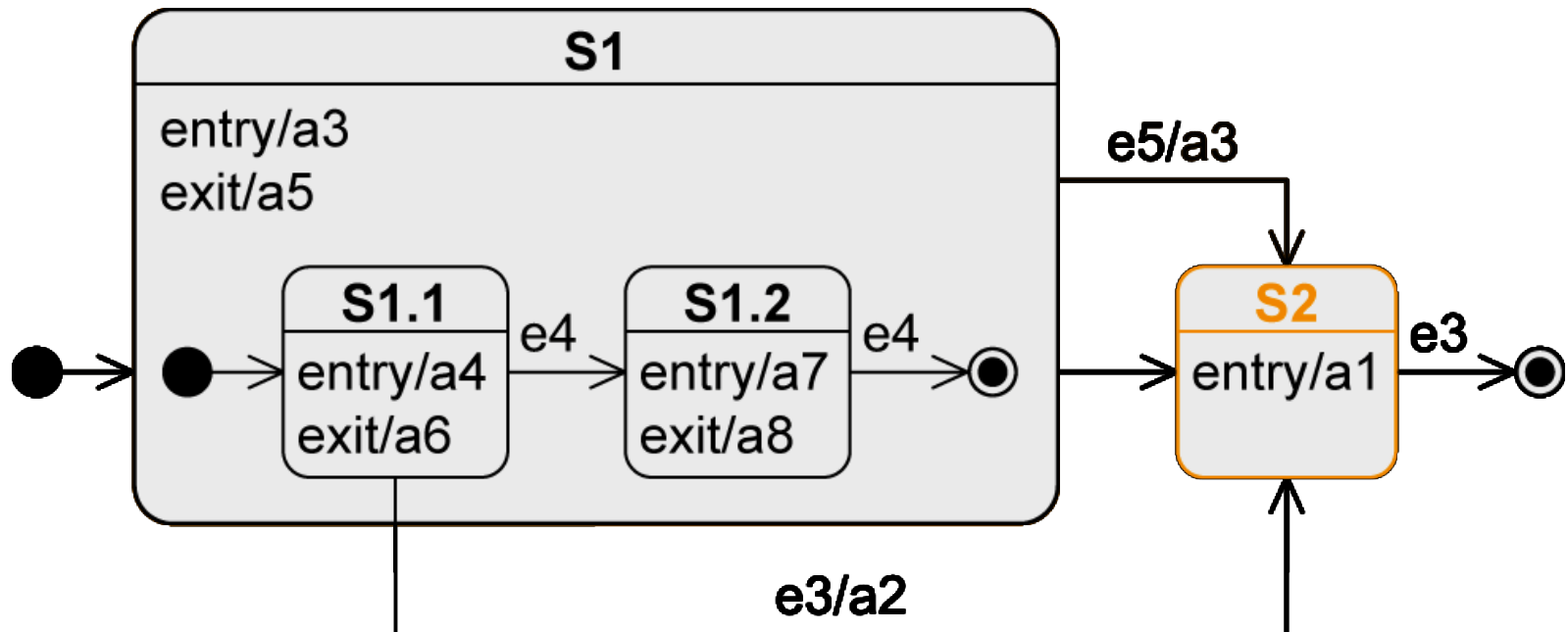


Exiting from a composite state (2/3)

- Transition from the composite state

No matter which substate of S1 is active, as soon as e5 occurs, the system changes to S2

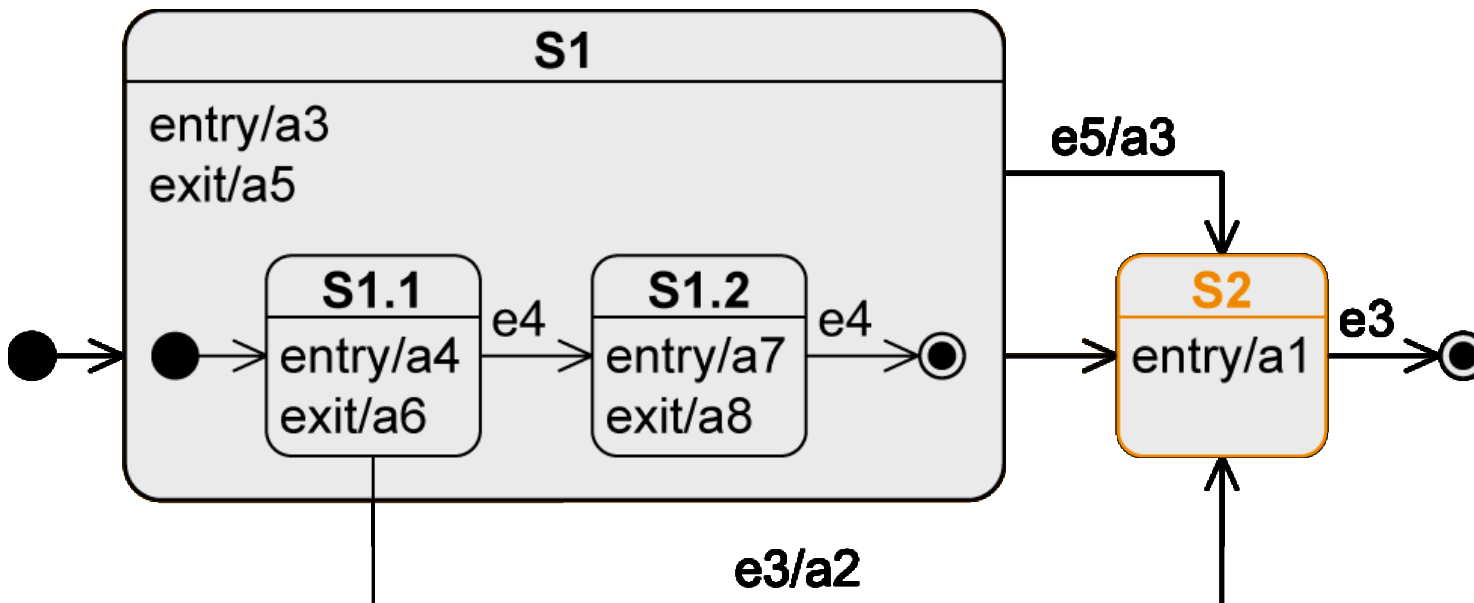
Event	State	Executed Activities
„Beginning“	S1/S1.1	a3-a4
e5	S2	a6-a5-a3-a1



Exiting from a composite state (3/3)

- Completion transition from the composite state

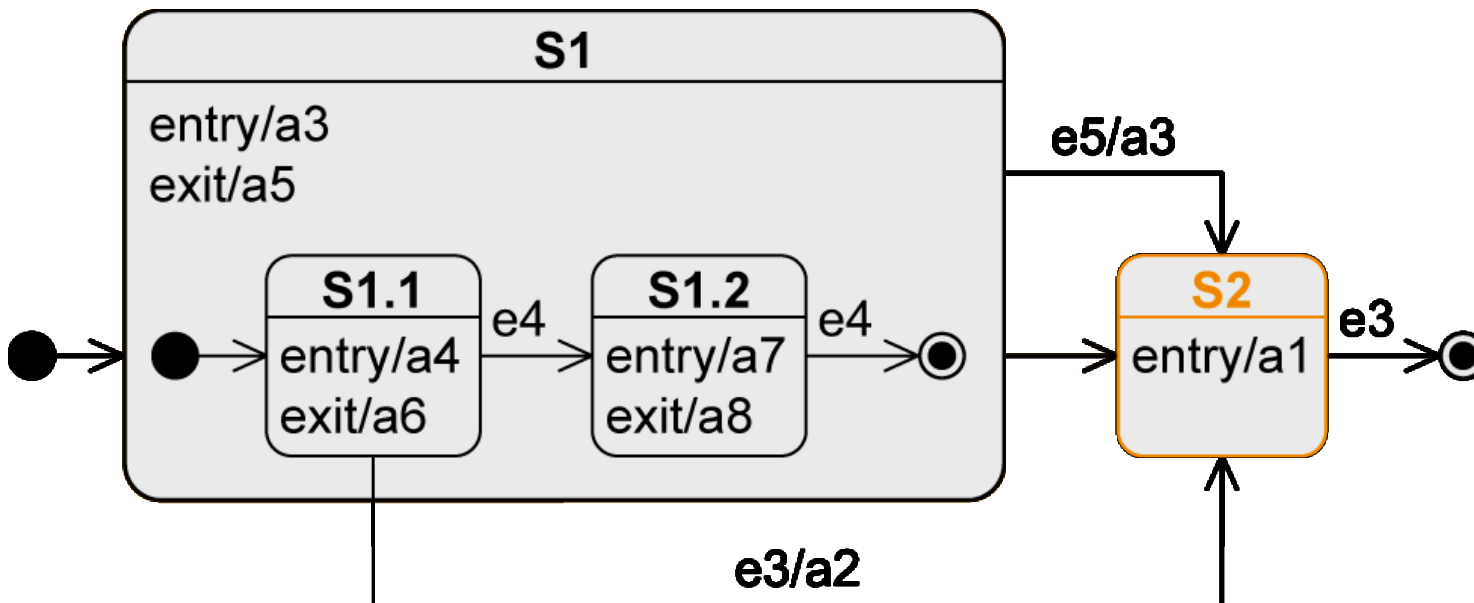
Event	State	Executed Activities
„Beginning“	...	...
e4	...	...
e4	...	...



Exiting from a composite state (3/3)

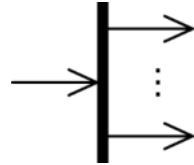
- Completion transition from the composite state

Event	State	Executed Activities
„Beginning“	S1/S1.1	a3-a4
e4	S1/S1.2	a6-a7
e4	S2	a8-a5-a1



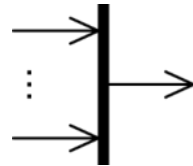
Parallelization and synchronization node

Parallelization node



- Pseudo state
- **Splits** the control flow into multiple concurrent flows
- 1 incoming edge
- At least 2 outgoing edges

Synchronization node

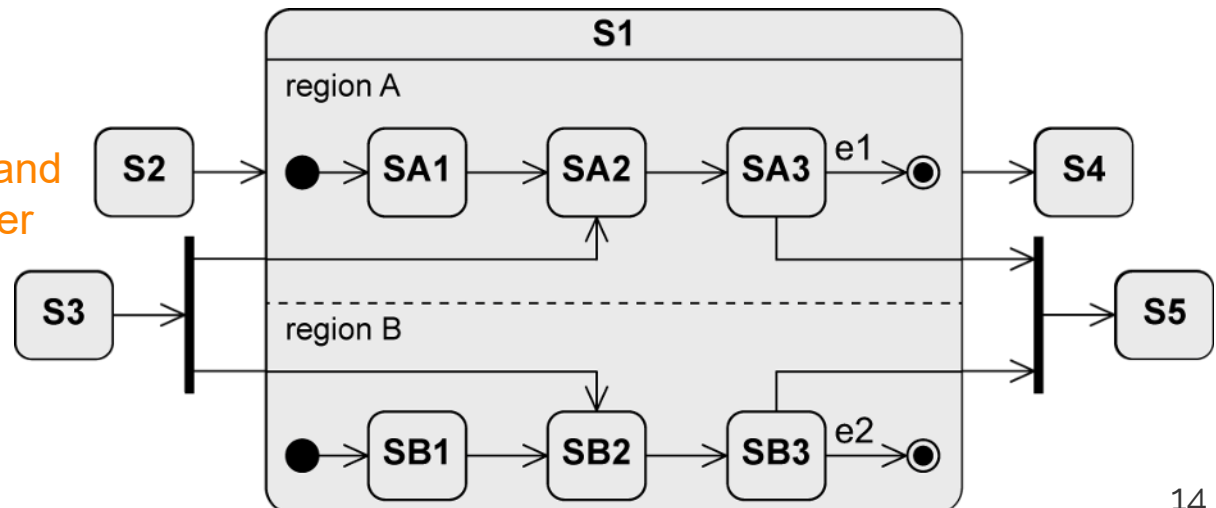


- Pseudo state
- **Merges** multiple concurrent flows
- At least 2 incoming edges
- 1 outgoing edge

Orthogonal state

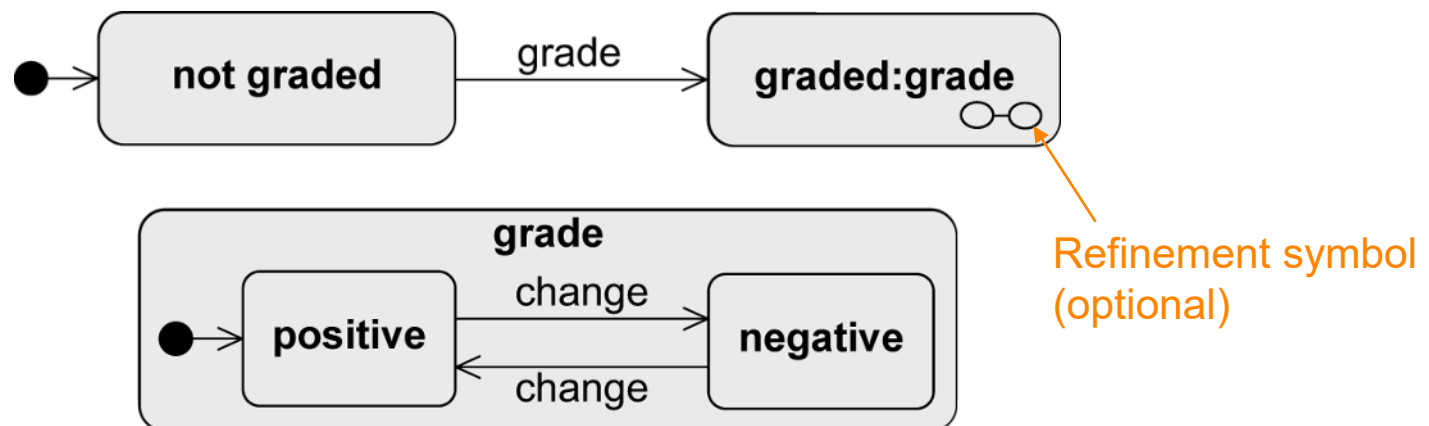
- Composite state is divided into two or more **regions** separated by a dashed line
- One state of each region is always active at any point in time, i.e., **concurrent substates**
- **Entry**: transition to the boundary of the orthogonal state activates the initial states of all regions
- **Exit**: final state must be reached in all regions to trigger completion event

You can use parallelization and synchronization node to enter different substates



Submachine state (SMS)

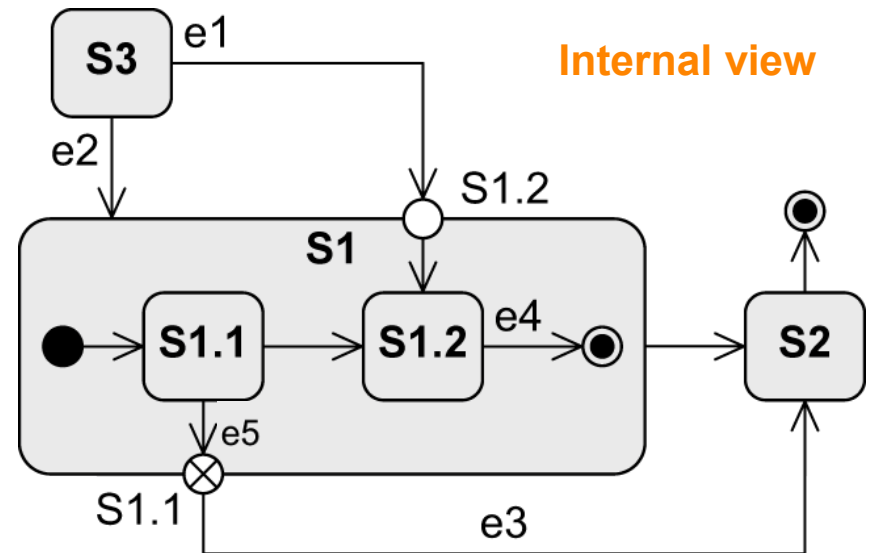
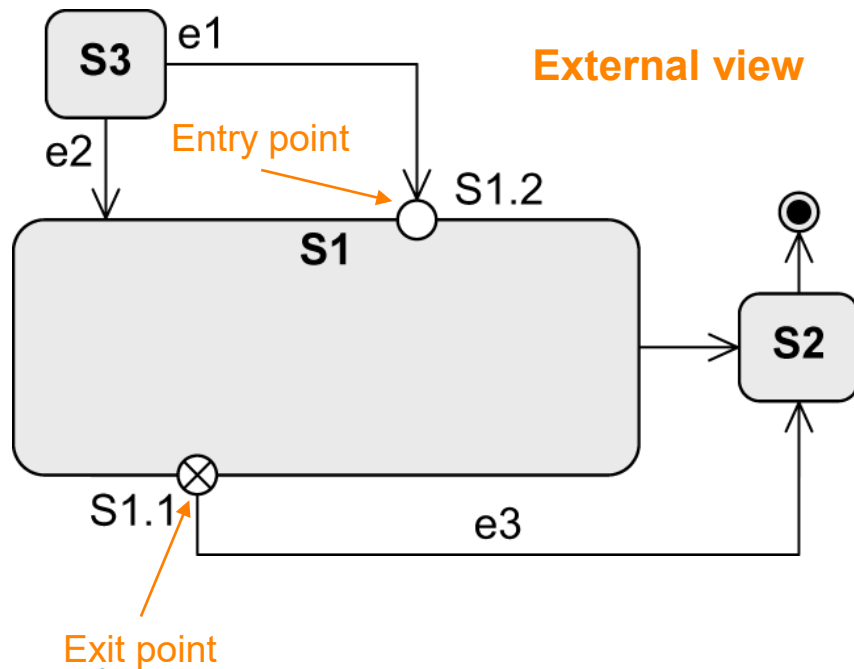
- Helpful for reusing parts of state machine diagrams in other state machine diagrams
- Notation: **stateName : submachineStateName**
- As soon as the submachine state is activated, the behavior of the submachine is executed
- Similar to calling a black-box method in Java



Entry and exit points

■ Encapsulation mechanism

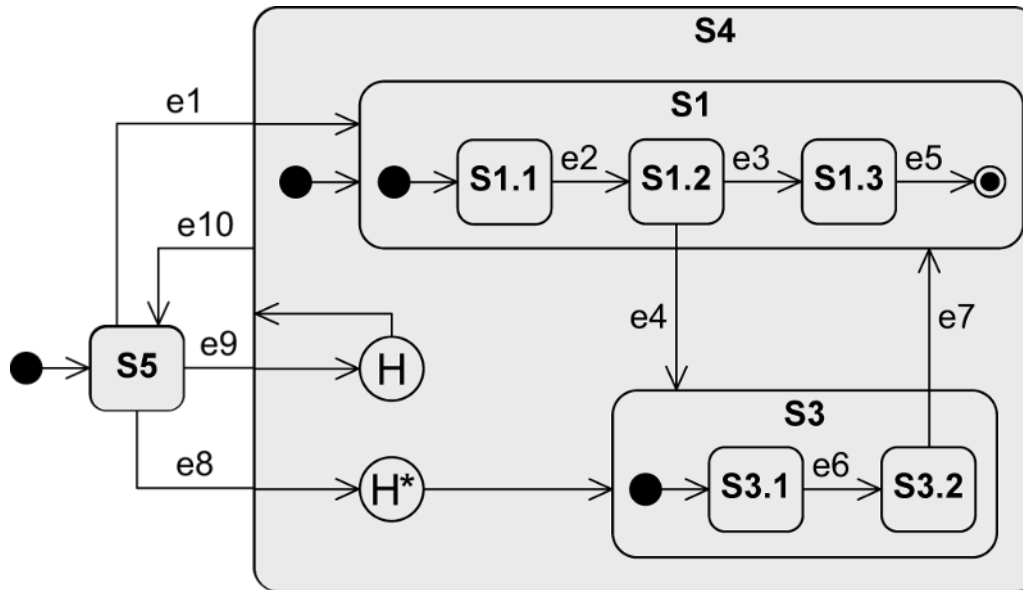
- A composite state shall be entered or exited via a state other than the initial and final states
- The external transition **does not know** the structure of the composite state



History state

- Remembers which substate of a composite state was the **last active** one
- Activates the “old” substate and all entry activities are conducted sequentially from the outside to the inside of the composite state
- Exactly one outgoing edge of the history state points to a substate. It is used if:
 - the composite state was never active before
 - OR
 - the composite state was exited via the final state
- Shallow vs. deep history state

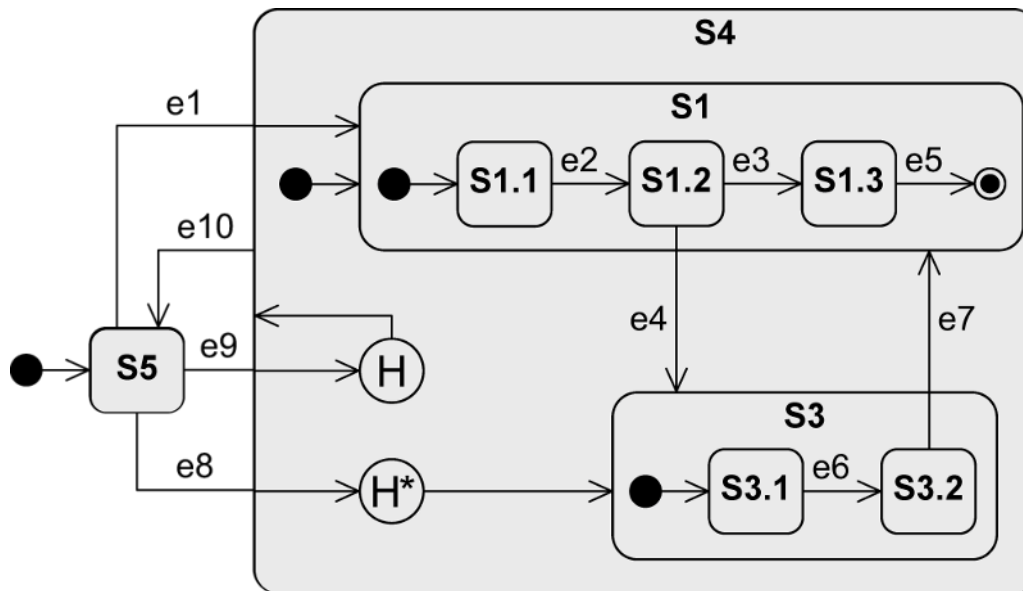
Example: history state (1/4)



Event	State
„Beginning“	S5
e1	S4/S1/S1.1
e2	S1.2
e10	S5
e9	(H→) S1/S1.1

H **Shallow history** state restores the last active state that is at the same level as the history state

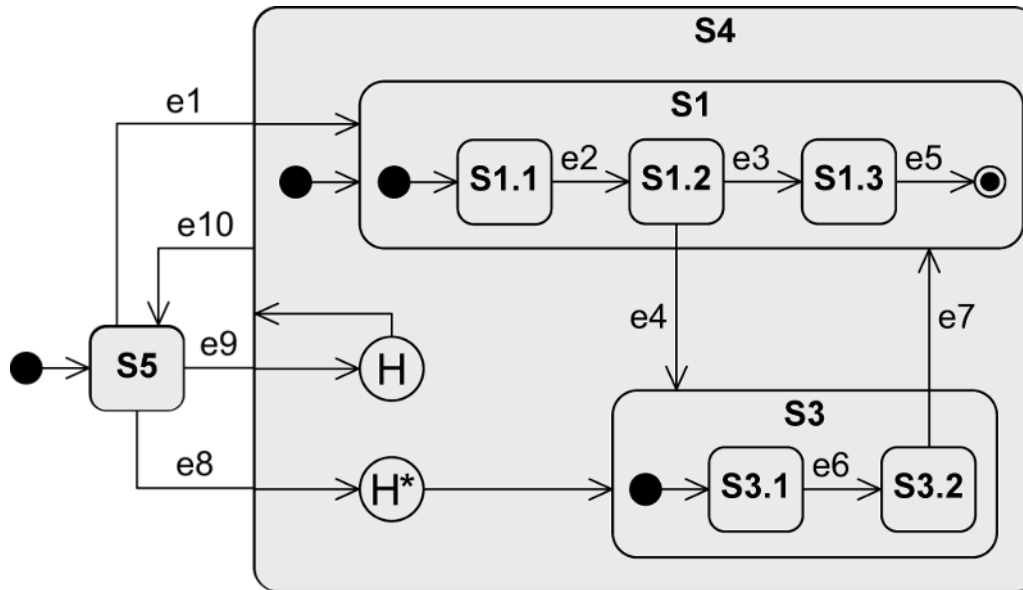
Example: history state (2/4)



Event	State
„Beginning“	S5
e1	S4/S1/S1.1
e2	S1.2
e10	S5
e8	(H* →) S1/S1.2

H* **Deep history** state restores the last active substate over the entire nesting depth

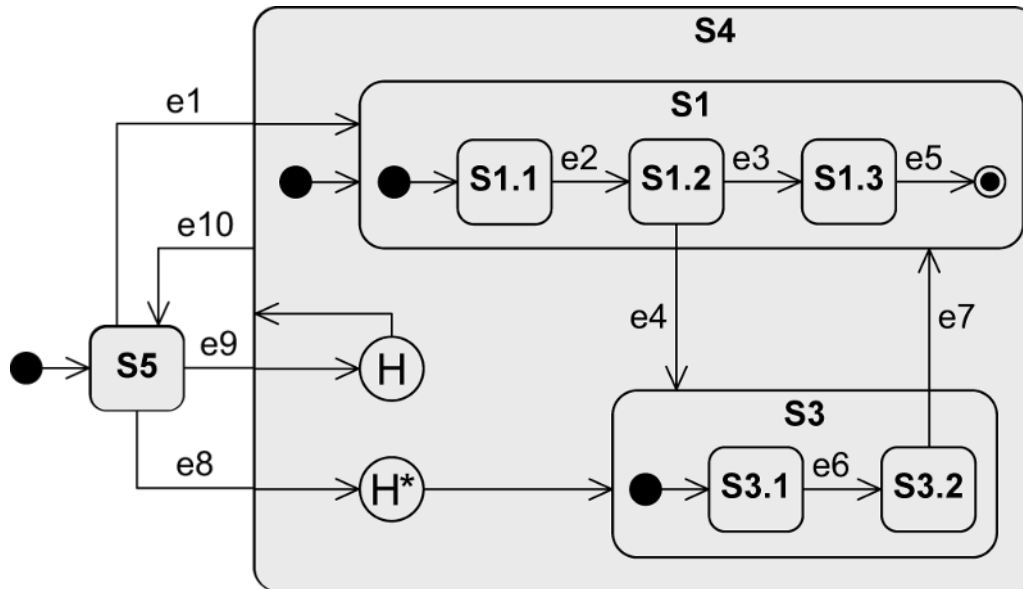
Example: history state (3/4)



Event	State
„Beginning“	S5
e9	(H→) S4/S1/S1.1

If no history exists, follow the outgoing transition.

Example: history state (4/4)



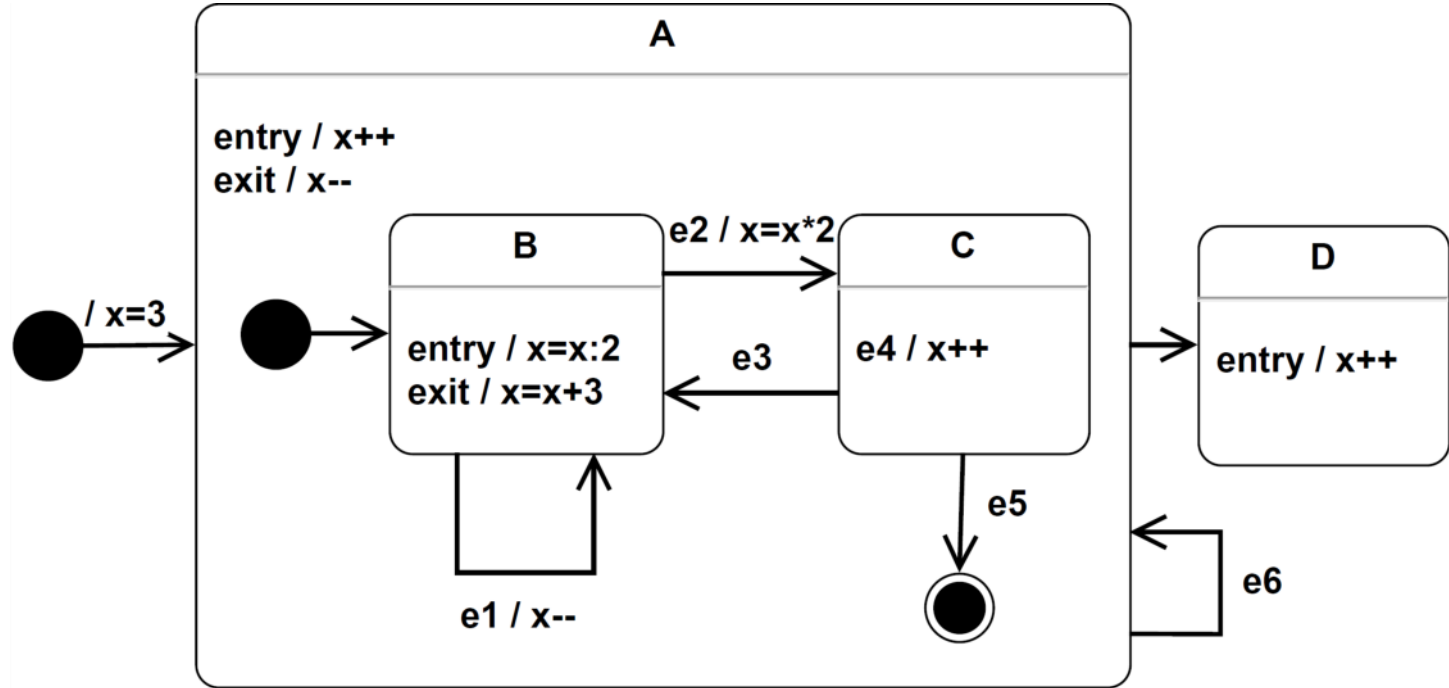
Event	State
„Beginning“	S5
e8	(H*→) S3/S3.1

If no history exists, follow the outgoing transition.

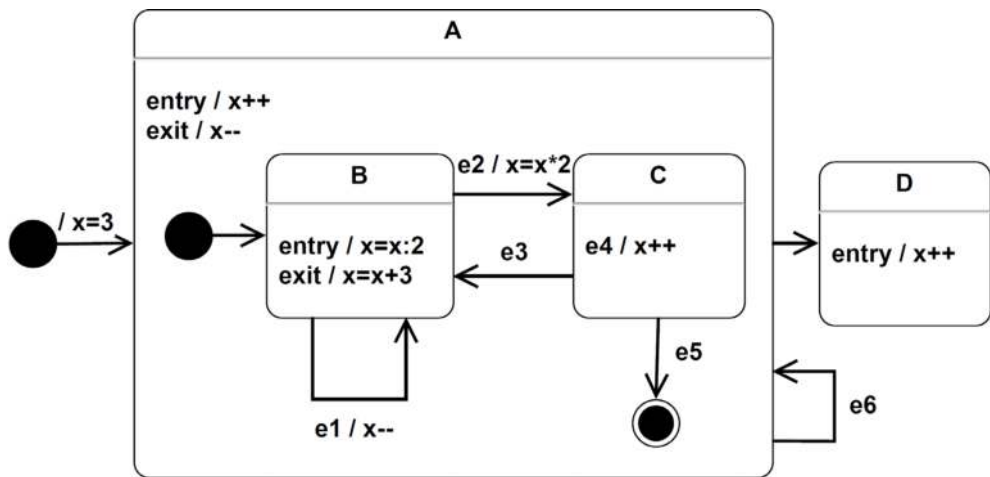
Exercise

Event sequence: e1, e2, e4, e4, e3, e1

What is the final value of x?

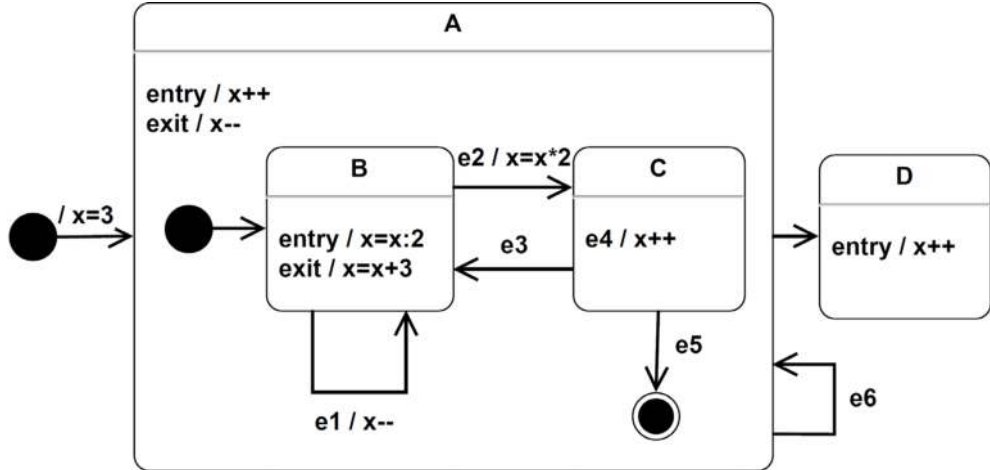


Solution



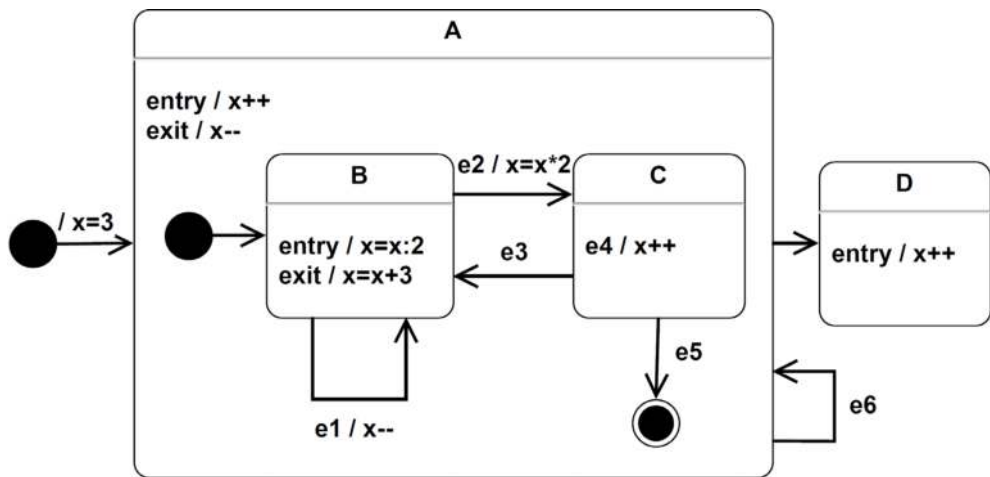
Event	State	Comment	x
		Start	3
e1			
e2			
e4			
e4			
e3			
e1			

Solution



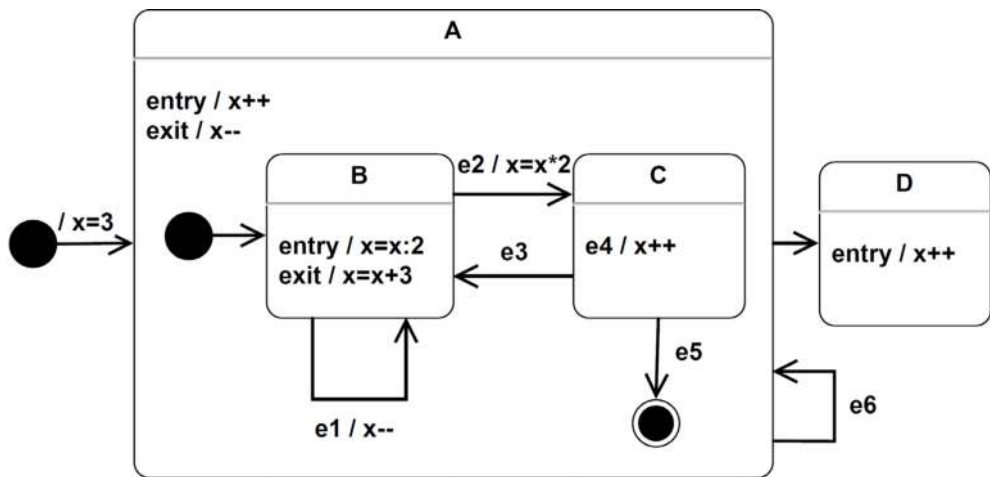
Event	State	Comment	x
		Start	3
	A	Entry of A	4
e1			
e2			
e4			
e4			
e3			
e1			

Solution



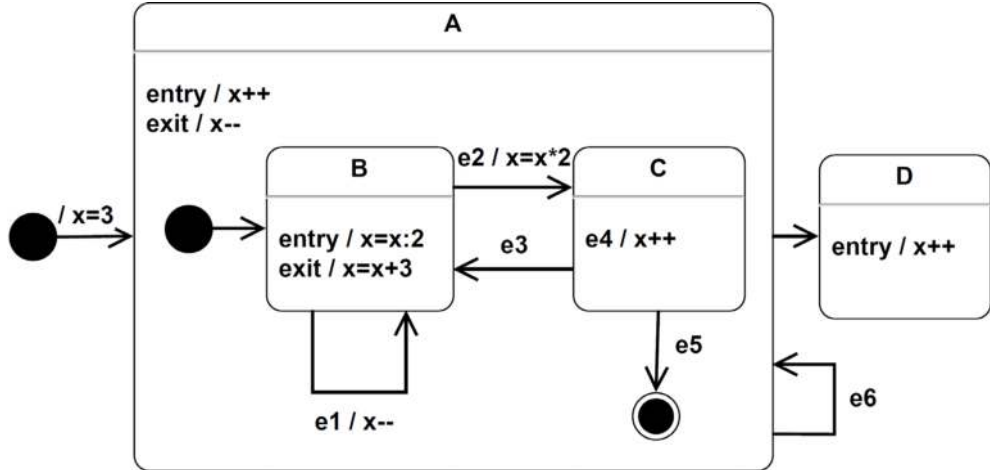
Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1			
e2			
e4			
e4			
e3			
e1			

Solution



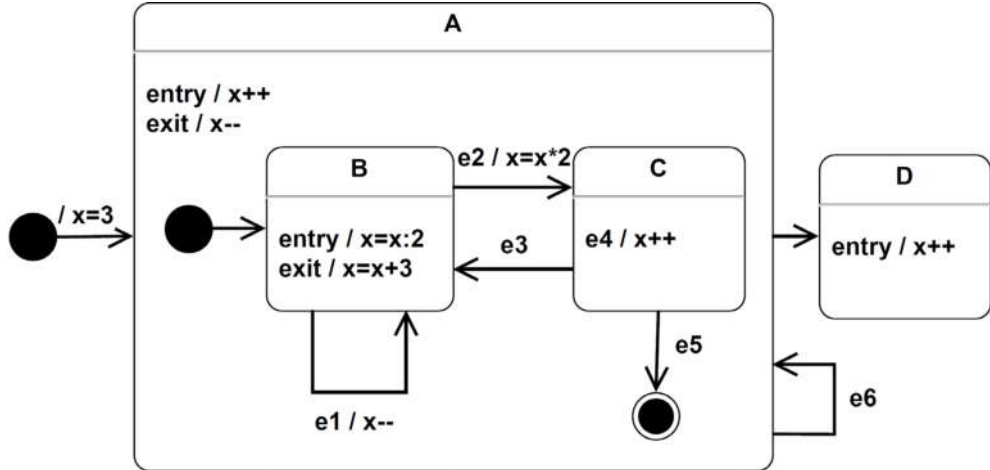
Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1	A/B	Exit of B	5
		x-- (self-transition)	4
		Entry of B	2
e2			
e4			
e4			
e3			
e1			

Solution



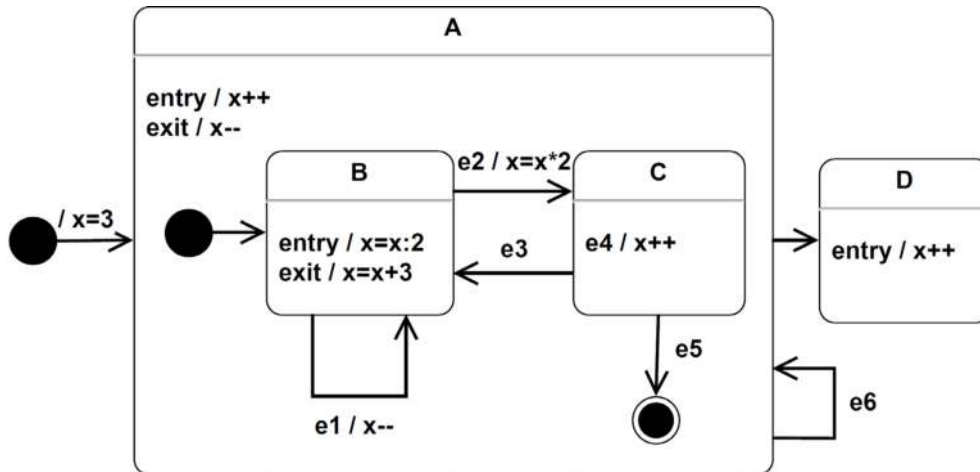
Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1	A/B	Exit of B	5
		x-- (self-transition)	4
		Entry of B	2
e2	A/C	Exit of B x=x*2	5 10
e4			
e4			
e3			
e1			

Solution



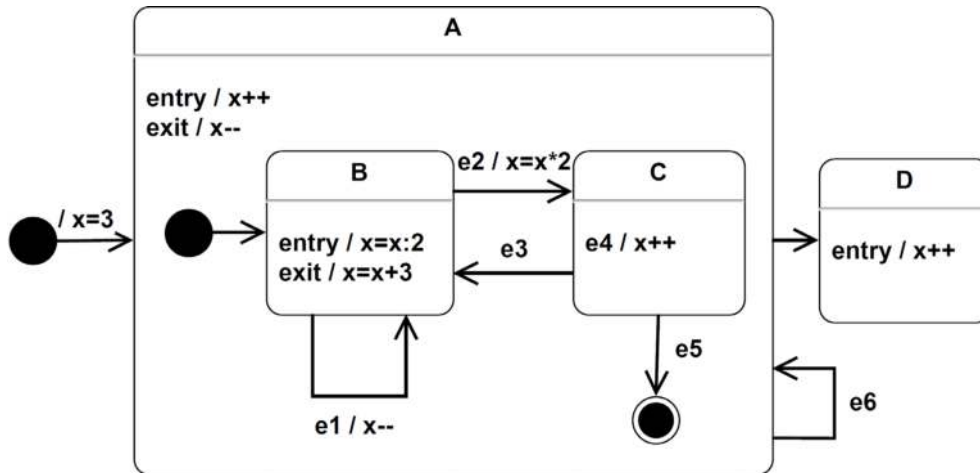
Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1	A/B	Exit of B	5
		x-- (self-transition)	4
		Entry of B	2
e2	A/C	Exit of B x=x*2	5 10
e4		x++ in C	11
e4		x++ in C	12
e3			
e1			

Solution



Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1	A/B	Exit of B $x--$ (self-transition) Entry of B	5 4 2
e2	A/C	Exit of B $x=x*2$	5 10
e4		$x++$ in C	11
e4		$x++$ in C	12
e3	A/B	Entry of B	6
e1			

Solution

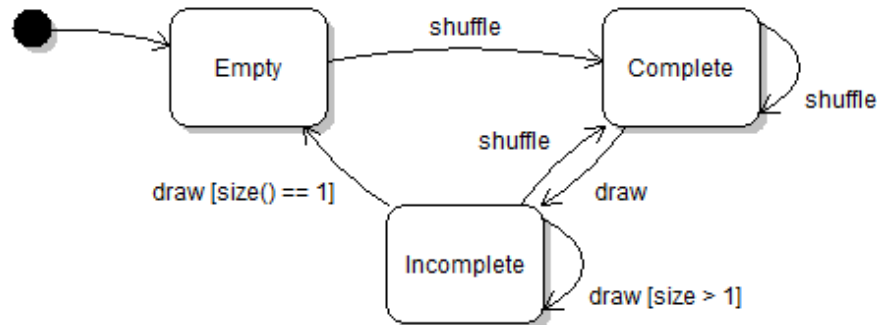


Event	State	Comment	x
		Start	3
	A	Entry of A	4
	A/B	Entry of B	2
e1	A/B	Exit of B	5
		x-- (self-transition)	4
		Entry of B	2
e2	A/C	Exit of B x=x*2	5 10
e4		x++ in C	11
e4		x++ in C	12
e3	A/B	Entry of B	6
e1		Exit of B	9
		x-- (self-transition)	8
		Entry of B	4

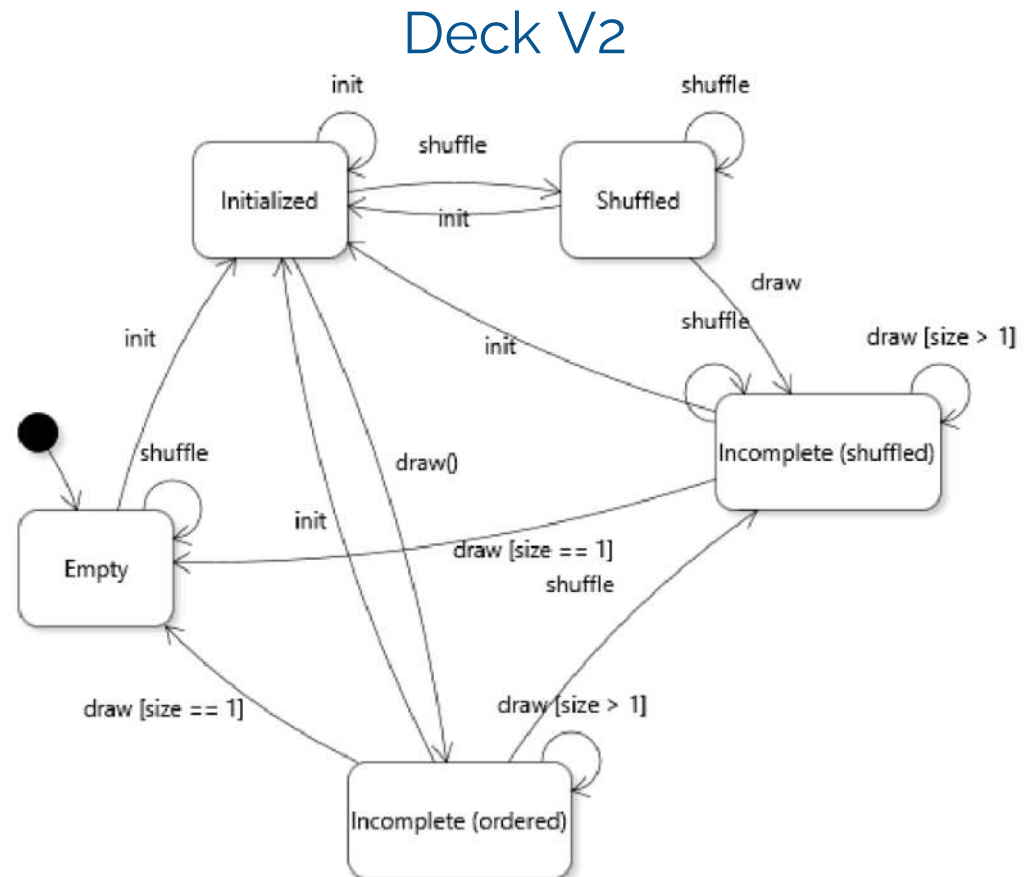
Design principles for state machines

- Minimize the state space of your objects
 - Put only the states that are strictly necessary for your object to satisfy their responsibilities
 - Objects with complex abstract state spaces are:
 - Difficult to use (i.e., to call their methods)
 - Difficult to test
 - Difficult to implement
- Avoid “speculative generalization”
 - “*What if we need this one day?*” may be a dangerous question
 - Less understandable code
 - More test cases to write
 - More sources of bugs
 - YAGNI principle: “You aren't gonna need it!”

Example of speculative generalization: a deck of cards



Deck V1



Deck V2

Key takeaways

- Design principles:
 - SRP
 - Encapsulation & immutability
 - Avoid complexity
- Your models are starting to move now!
- State machine used for modelling the **internal states** of objects in your system
- You may have **a state machine for each important class** in your class diagrams
 - You may need fewer, e.g., basic data structures are usually passive entities → no state machines for them
 - You model only the **key states** of the objects, not everything

Readings

- UML@Classroom: An Introduction to Object-Oriented Modeling" – Chapter 5
- Introduction to Software Design with Java – Chapters 2 and 4.1, 4.2, 4.3, 4.4
- Engineering Software Products, Chapter 8.1
- A philosophy of Software Design, Chapters 7, 8, 9
- [optional] Learning UML 2.0 – chapter 14